Wave and Vibration (Supplementary Problems)

1. A sound wave has a frequency of 192 Hz and travels the length of a football field, 91.4 m, in 0.271 s.
a. What is the speed of the wave?
b. What is the wavelength of the wave?
c. What is the period of the wave?
d. If the frequency was changed to 442 Hz, what would be the new wavelength and period?

2. A sound wave produced by a clock chime is heard 515 m away 1.50 s later.
a. What is the speed of sound of the clock's chime in air?
b. The sound wave has a frequency of 436 Hz. What is the period of the wave?
c. What is the wave's wavelength?

3. A hiker shouts toward a vertical cliff 465 m away. The echo is heard 2.75 s later.
a. What is the speed of sound of the hiker's voice in air?
b. The wavelength of the sound is 0.750 m. What is its frequency?
c. What is the period of the wave?

4. If you want to increase the wavelength of waves in a rope, should you shake it at a higher or lower frequency?

5. What is the speed of a periodic wave disturbance that has a frequency of 3.50 Hz and a wavelength of 0.700 m?

6. The speed of a transverse wave in a string is 15.0 m/s. If a source produces a disturbance that has a frequency of 6.00 Hz, what is its wavelength?

7. Five pulses are generated every 0.100 s in a tank of water. What is the speed of propagation of the wave if the wavelength of the surface wave is 1.20 cm?

8. A periodic longitudinal wave that has a frequency of 20.0 Hz travels along a coil spring. If the distance between successive compressions is 0.600 m, what is the speed of the wave?

9. A spring stretches by 0.12 m when some apples weighing 3.2 N are suspended from it, as shown in Figure 14-20. What is the spring constant of the spring?

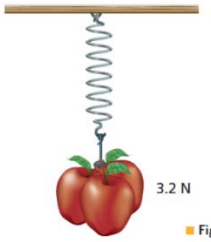


Figure 14-20

10. Car Shocks Each of the coil springs of a car has a spring constant of 25,000 N/m. How much is each spring compressed if it supports one-fourth of the car's 12,000-N weight?

11. How much potential energy is stored in a spring with a spring constant of 27 N/m if it is stretched by 16 cm?

12. Rocket Launcher A toy rocket-launcher contains a spring with a spring constant of 35 N/m. How far must the spring be compressed to store 1.5 J of energy?

13. Force-versus-length data for a spring are plotted on the graph in Figure 14-21.
a. What is the spring constant of the spring?
b. What is the energy stored in the spring when it is stretched to a length of 0.50 m?

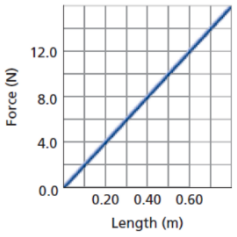


Figure 14-21

14. How long must a pendulum be to have a period of 2.3 s on the Moon, where g = 1.6 m/s^2?

15. Building Motion The Sears Tower in Chicago, shown in Figure 14-22, sways back and forth in the wind with a frequency of about 0.12 Hz. What is its period of vibration?



Figure 14-22

16. Ocean Waves An ocean wave has a length of 12.0 m. A wave passes a fixed location every 3.0 s. What is the speed of the wave?

17. Water waves in a shallow dish are 6.0-cm long. At one point, the water moves up and down at a rate of 4.8 oscillations/s.
a. What is the speed of the water waves?
b. What is the period of the water waves?

18. Water waves in a lake travel 3.4 m in 1.8 s. The period of oscillation is 1.1 s. a. What is the speed of the water waves? b. What is their wavelength?

19. Sonar A sonar signal of frequency 1.00×10^{6} Hz has a wavelength of 1.50 mm in water.
a. What is the speed of the signal in water?
b. What is its period in water?
c. What is its period in air?

20. A sound wave of wavelength 0.60 m and a velocity of 330 m/s is produced for 0.50 s.

a. What is the frequency of the wave?

b. How many complete waves are emitted in this time interval?

c. After 0.50 s, how far is the front of the wave from the source of the sound?

21. The speed of sound in water is 1498 m/s. A sonar signal is sent straight down from a ship at a point just below the water surface, and 1.80 s later, the reflected signal is detected. How deep is the water?

22. Pepe and Alfredo are resting on an offshore raft after a swim. They estimate that 3.0 m separates a trough and an adjacent crest of each surface wave on the lake. They count 12 crests that pass by the raft in 20.0 s. Calculate how fast the waves are moving.

23. Earthquakes The velocity of the transverse waves produced by an earthquake is 8.9 km/s, and that of the longitudinal waves is 5.1 km/s. A seismograph records the arrival of the transverse waves 68 s before the arrival of the longitudinal waves. How far away is the earthquake?

24. If you slosh the water in a bathtub at the correct frequency, the water rises first at one end and then at the other. Suppose you can make a standing wave in a 150-cm-long tub with a frequency of 0.30 Hz. What is the velocity of the water wave?

25. Guitars The wave speed in a guitar string is 265 m/s. The length of the string is 63 cm. You pluck the center of the string by pulling it up and letting go. Pulses move in both directions and are reflected off the ends of the string.

a. How long does it take for the pulse to move to the string end and return to the center?

b. When the pulses return, is the string above or below its resting location?

c. If you plucked the string 15 cm from one end of the string, where would the two pulses meet?

26. What is the period of a pendulum with a length of 1.4 m?

27. The frequency of yellow light is 5.1×10^{14} Hz. Find the wavelength of yellow light. The speed of light is 3.00×10^{8} m/s.

28. Radio Wave AM-radio signals are broadcast at frequencies between 550 kHz (kilohertz) and 1600 kHz and travel 3.0×10^8 m/s.

a. What is the range of wavelengths for these signals?

b. FM frequencies range between 88 MHz (megahertz) and 108 MHz and travel at the same speed. What is the range of FM wavelengths?

29. You are floating just offshore at the beach. Even though the waves are steadily moving in toward the beach, you don't move any closer to the beach.

a. What type of wave are you experiencing as you float in the water?

b. Explain why the energy in the wave does not move you closer to shore.

c. In the course of 15 s you count ten waves that pass you. What is the period of the waves?

d. What is the frequency of the waves?

e. You estimate that the wave crests are 3 m apart. What is the velocity of the waves?

f. After returning to the beach, you learn that the waves are moving at 1.8 m/s. What is the actual wavelength of the waves?

30. Bungee Jumper A high-altitude bungee jumper jumps from a hot-air balloon using a 540-m-bungee cord. When the jump is complete and the jumper is just suspended from the cord, it is stretched 1710 m. What is the spring constant of the bungee cord if the jumper has a mass of 68 kg?

31. The time needed for a water wave to change from the equilibrium level to the crest is 0.18 s.
a. What fraction of a wavelength is this?
b. What is the period of the wave?
c. What is the frequency of the wave?

32. When a 225-g mass is hung from a spring, the spring stretches 9.4 cm. The spring and mass then are pulled 8.0 cm from this new equilibrium position and released. Find the spring constant of the spring and the maximum speed of the mass.

33. Amusement Ride You notice that your favorite amusement-park ride seems bigger. The ride consists of a carriage that is attached to a structure so it swings like a pendulum. You remember that the carriage used to swing from one position to another and back again eight times in exactly 1 min. Now it only swings six times in 1 min. Give your answers to the following questions to two significant digits.
a. What was the original period of the ride?
b. What is the new period of the ride?
c. What is the new frequency?
d. How much longer is the arm supporting the carriage on the larger ride?
e. If the park owners wanted to double the period of the ride, what percentage increase would need to be made to the length of the pendulum?

34. Clocks The speed at which a grandfather clock runs is controlled by a swinging pendulum. a. If you find that the clock loses time each day, what adjustment would you need to make to the pendulum so it will keep better time? b. If the pendulum currently is 15.0 cm, by how much would you need to change the length to make the period lessen by 0.0400 s?

35. Bridge Swinging In the summer over the New River in West Virginia, several teens swing from bridges with ropes, then drop into the river after a few swings back and forth.

a. If Pam is using a 10.0-m length of rope, how long will it take her to reach the peak of her swing at the other end of the bridge?

b. If Mike has a mass that is 20 kg more than Pam, how would you expect the period of his swing to differ from Pam's?

c. At what point in the swing is KE at a maximum?

d. At what point in the swing is PE at a maximum?

e. At what point in the swing is KE at a minimum?

f. At what point in the swing is PE at a minimum?

36. You have a mechanical fish scale that is made with a spring that compresses when weight is added to a hook attached below the scale. Unfortunately, the calibrations have completely worn off of the scale. However, you have one known mass of 500.0 g that displaces the spring 2.0 cm.

a. What is the spring constant for the spring?

b. If a fish displaces the spring 4.5 cm, what is the mass of the fish?

37. Car Springs When you add a 45-kg load to the trunk of a new small car, the two rear springs compress an additional 1.0 cm. a. What is the spring constant for each of the springs? b. How much additional potential energy is stored in each of the car springs after loading the trunk?

38. The velocity of a wave on a string depends on how tightly the string is stretched, and on the mass per unit length of the string. If F_T is the tension in the string, and μ is the mass/unit length, then the velocity, v, can be determined by the following equation. $v = \sqrt{\frac{F_T}{\mu}}$ piece of string 5.30-m long has a mass of 15.0 g. What must the tension in the string be to make the wavelength of a 125-Hz wave 120.0 cm?

39. Analyze and Conclude A 20-N force is required to stretch a spring by 0.5 m.

a. What is the spring constant?

b. How much energy is stored in the spring?

c. Why isn't the work done to stretch the spring equal to the force times the distance, or 10 J?

40. Make and Use Graphs Several weights were suspended from a spring, and the resulting extensions of the spring were measured. Table 14-1 shows the collected data.

Table 14-1
Weights on a Spring

Force, F (N)	Extension, x (m)
2.5	0.12
5.0	0.26
7.5	0.35
10.0	0.50
12.5	0.60
15.0	0.71

41. What is the value of the spring constant of a spring with a potential energy of 6.67 J when it's stretched 247 mm?

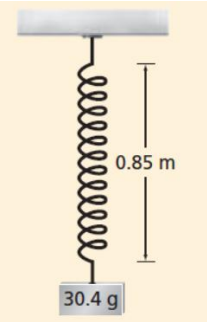
a) 70.2 N/m b) 71.1 N/m c) 142 N/m d) 284 N/m

42. What is the force acting on a spring with a spring constant of 275 N/m that is stretched 14.3 cm?

a) 2.81 N b) 19.2 N c) 39.3 N d) 3.93×10^3 N

43. A mass stretches a spring as it hangs from the spring. What is the spring constant?

a) 0.25 N/m
b) 0.35 N/m
c) 26 N/m
d) 3.5×10^2 N/m



44. A spring with a spring constant of 350 N/m pulls a door closed. How much work is done as the spring pulls the door a constant velocity from an 85.0-cm stretch to a 5.0-cm stretch?

a) 112 N·m b) 130 J c) 224 N·m d) 1.12×10^3 J

45. What is the correct rearrangement of the formula for the period of a pendulum to find the length of the pendulum?

a) $l = 4\pi t^2/g$ b) $l = T^2/4\pi^2$ c) $l = gt^2/4\pi^2$ d) $l = Tt/2\pi$

46. What is the frequency of a wave with a period of 3 s?

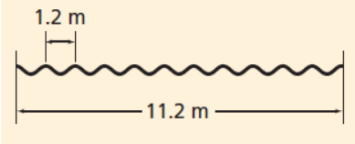
a) 0.3 Hz b) 1 Hz c) 3 Hz d) 1/3 Hz

47. Which option describes a standing wave?

	Waves	Direction	Medium
Ⓐ	Identical	Same	Same
Ⓑ	Nonidentical	Opposite	Different
Ⓒ	Identical	Opposite	Same
Ⓓ	Nonidentical	Same	Different

48. A 1.2-m wave travels 11.2 m to a wall and back again in 4 s. What is the frequency of the wave?

a) 0.2 Hz
b) 2 Hz
c) 5 Hz
d) 9 Hz



49. What is the length of a pendulum that has a period of 4.89 s?

a) 5.94 m b) 11.9 m c) 24.0 m d) 37.3 m

50. Use dimensional analysis of the equation kx = mg to derive the units of k.

51. The period of a simple pendulum is 1s on Earth. When brought to a planet where g is one-tenth that on Earth, what is its new period?

52. A man enters a tall tower, needing to know its height. He notes that a long pendulum extends from the ceiling almost to the floor and that its period is 15.5 s.

a. How tall is the tower?

b. If this pendulum is taken to the Moon, where the free-fall acceleration is 1.67 m/s^2, what is the period there?

53. A simple pendulum has a length of 50.0 cm and makes 80.0 complete oscillations in 2.00 min.
Find;

a. the period of the pendulum

b. the value of g at the location of the pendulum

54. The sinusoidal wave shown in Figure P13.41 is traveling in the positive x-direction and has a frequency of 18.0 Hz. Find the amplitude, wavelength, period, and speed of the wave.

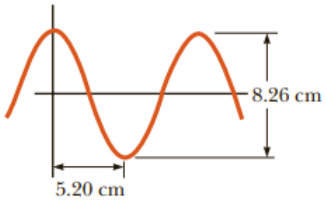


Figure P13.41

55. A spring oriented vertically is attached to a hard horizontal surface as in Figure P13.2. The spring has a force constant of 1.46 kN/m. How much is the spring compressed when an object of mass m = 2.30 kg is placed on top of the spring and the system is at rest?

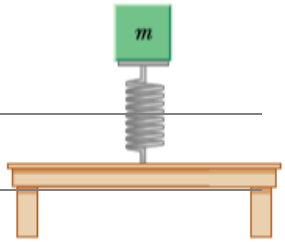


Figure P13.2

56. A 0.75 m spring is fixed in the wall. Calculate the elastic potential energy stored in a spring, when it is compressed to a length of 0.55 m. (Take the value of spring constant, k = 230 N/m)

57. A force of 6.0 N is applied to a metal wire, which extends by 0.040 m. Calculate the stiffness constant of the wire.

58. A spring of original length 40 mm is suspended from a stand and clamp. A mass of 1.5 kg Is attached to the bottom and the spring extends to a total length of 80 mm as shown in **Figure (a).**

a. Calculate the spring constant of the spring in $N.m^{-1}$

b. The mass is now pulled down, extending the spring a further 30 mm as shown in **Figure (b).**

i. Calculate the unbalanced force acting on the mass just after its release.

ii. Calculate the acceleration of oscillation of the mass just after its release.

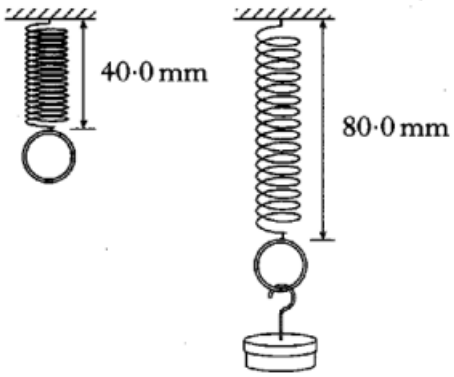


Figure (a)

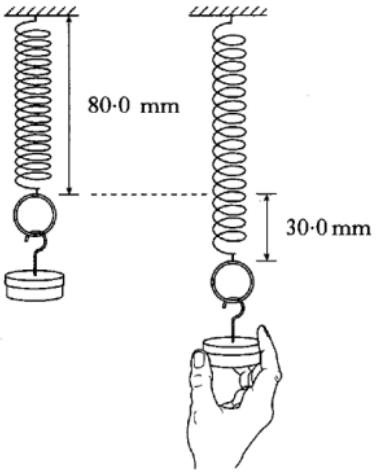
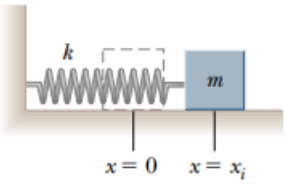


Figure (b)

59. 0.500 kg object connected to a light spring with a spring constant of 40.0 N/m oscillates on a frictionless horizontal surface.



a. Calculate the total energy of the system and the maximum speed of the object if the amplitude of the motion is $x_i = 3.0$ cm.

b. What is the velocity of the object when the displacement is 2.00 cm?

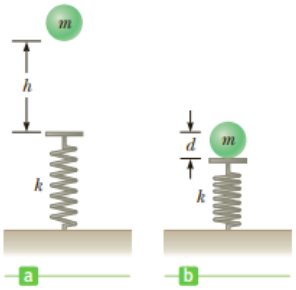
c. Compute the kinetic and potential energies of the system when the displacement is 2.00 cm.

d. Find the distance x where the kinetic energy will equal to potential energy.

60. A ball of mass m=2.0 kg is released from rest at a height h =20.0 cm above a light vertical spring of force constant k as in **Figure (a)**. The ball strikes the top of the spring and compresses it a distance d=5.0 cm as in **Figure (b)**. Neglecting any energy losses during the collision, find

a. the speed of the ball just as it touches the spring.

b. the force constant of the spring. Given: g = 10 m/s^2.

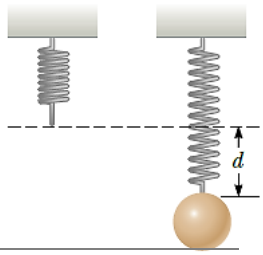


61. A wave is described by $y = 0.0200\sin(kx - \omega t)$, where $k = 2.11 rad/m, \omega = 3.62 rad/s$, x and y are in meters, and t is in seconds. Determine (a) the amplitude, (b) the wavelength, (c) the frequency, and (d) the speed of wave.

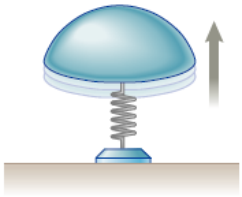
62. In an engine, a piston oscillates with simple harmonic motion so that its position varies according to expression $x = 5.00\cos\left(2t + \frac{\pi}{6}\right)$, where x is in centimeters and t is in seconds. At $t = 0$, find (a) the position of the particles, (b) its velocity, and (c) its acceleration. Find (d) the period and (e) the amplitude of the motion.

63. A uniform cord has a mass of $0.3kg$ and length of $6m$, the cord passes over pulley and support a $2kg$ object. (a) What will be the speed of wave? (b) What will be the time for wave to travel the distance $6m$?

64. A spring is hung from a ceiling, and an object attached to its lower end stretches the spring by a distance $d = 5.00$ cm from its unstretched position when the system is in equilibrium as in Figure here. If the spring constant is 47.5 N/m, determine the mass of the object.



65. A child's toy consists of a piece of plastic attached to a spring (Fig. below). The spring is compressed against the floor a distance of 2.00 cm, and the toy is released. If the toy has a mass of 100 g and rises to a maximum height of 60.0 cm, estimate the force constant of the spring.



66. An automobile having a mass of 1 000 kg is driven into a brick wall in a safety test. The bumper behaves like a spring with constant 5.00×10^6 N/m and is compressed 3.16 cm as the car is brought to rest. What was the speed of the car before impact, assuming no energy is lost in the collision with the wall?

67. After landing on an unfamiliar planet, a space explorer constructs a simple pendulum of length 50 cm. She finds that the pendulum makes 100 complete swings in 250s. What is the value of g on this planet? (use $\pi^2 = 10$)

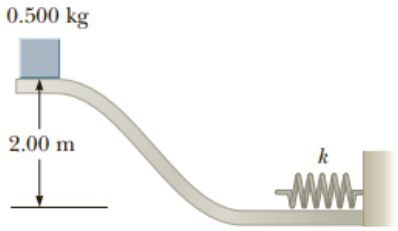
68. What is the period of a block that has an acceleration of -3 m/s^2 when the block is 0.30 m from its equilibrium position?

69. A simple pendulum of length L_1 has a time period of 6s and another simple pendulum of length L_2 has a time period 8s. What is the time period of another pendulum of length $L_1 + L_2$?

70. A 2 kg mass on a spring has displacement as a function of time given by the equation $x(t) = 10\sin(8\pi t)\ (cm)$. Find

a. the time of one complete vibration,
b. the frequency of the motion,
c. the amplitude of the motion,
d. the spring constant,
e. the maximum speed of the mass,
f. the maximum acceleration of the mass,
g. the first time after $t = 0$ that the object reaches the position $x = 10$ cm,
h. the mechanical energy of the system.

71. 0.500 kg block is released from rest and slides down a frictionless track that begins 2.00 m above the horizontal, as shown in Figure below. At the bottom of the track, where the surface is horizontal, the block strikes and sticks to a light spring with a spring constant of 20.0 N/m. Find the maximum distance the spring is compressed.



72. What is the period of a block that has an acceleration of -3 m/s^2 when the block is 0.30 m from its equilibrium position?